V2Power Coupon Admin — Import 功能 UAT 验收测试报告

文档版本：v1.0 终版　　**测试日期：**2026-08-28　　**测试环境：**内网预发布（192.168.33.163）+ 生产联调（v2power.io）

测试对象：http://192.168.33.163/couponAdmin/admin/import　　**测试者：**Lois

架构：浏览器 → 192.168.33.163:80（nginx 托管 React SPA）→ /coupon_api 代理 → 192.168.33.100:8088（Spring Boot）；Admin 前端经设计直连 v2power.io 生产 API 管理生产券数据

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1. 测试结论

结论：Admin Import 核心功能全部通过，端到端联调闭环，建议发布。全新部署全量重测，覆盖真实浏览器运营流程、导入历史、接口性能、安全测试、以及 Import↔Redeem 生产联调。已测项无失败。剩余 3 个 P2 / 2 个 P3 改进项（服务端校验、CORS/鉴权、错误处理），不阻塞发布，建议后续迭代优化。

测试类别	用例数	通过	失败	观察	关键证据
真实浏览器运营流程	8	**8**	0	0	登录/导入/QR/Flyer/历史/公开页
导入历史 / localStorage	6	**6**	0	0	新增/持久化/清空/上限/Copy/ZIP
接口与性能	5	**5**	0	0	1000行8.03s/幂等/并发去重
安全测试	7	2	0	5	无鉴权/CORS全开/未校验格式
Import↔Redeem 端到端联调	1	**1**	0	0	生产兑换成功，邮件优惠码已收
合计	**27**	**22**	**0**	**5**	已测项通过率 100%，无失败

2. 环境与部署配置

项目	值
前端地址（Nginx）	http://192.168.33.163/couponAdmin（nginx/1.20.1，React + Vite SPA）
前端 Bundle	index-K0xbCKB_.js（430,979 B）/ index-7tdCxuTj.css（42,636 B），Last-Modified 2026-08-28 10:33:52
后端（163 代理）	http://192.168.33.163/coupon_api → 192.168.33.100:8088（Spring Boot）
生产 API（Admin 直连）	https://v2power.io（.env.production 的 VITE_API_TARGET，Admin UI 导入/下载经此写入生产库）
设计说明	Admin 后台部署于公司本地、用户端部署于亚马逊，Admin 前端硬编码生产 API 属有意设计，用于直接管理生产券数据，**非缺陷**
破图修复确认	GET /couponAdmin/v2power-logo.png → 200（842,267 B）；GET /couponAdmin/charger-cut.png → 200（319,462 B）

3. 后台路由清单（全量审计）

有效路由 7 条（均 200 可访问）：

· / （兑换页） · /success （兑换成功页） · /help （帮助页） · /privacy （隐私政策页）

· /admin/login （登录页） · /admin/import （导入页；/admin 自动重定向到此） · /admin/flyer/:id （Flyer 打印/预览页）

不存在 / 死代码：/admin/batches（批次列表）、/admin/batch/:id（批次详情）、/admin/print（打印）——源码 BatchesView/BatchDetailView/PrintView 为死代码，bundle 0 命中；后端 GET /admin/batches、GET /admin/batch/{id}、POST /admin/login 均返回 404（未实现）。

结论：批次/优惠券列表页、优惠券详情页、兑换记录页、系统设置/用户管理页均不存在（后端也未实现）。Admin 当前功能范围 = 导入 + Flyer 预览 + 本地历史。

4. 功能测试（真实浏览器用户故事）

4.1 故事 1：登录 → 进入导入页 — PASS

访问 /admin/login，显示 V2Power ADMIN 登录页，Password 输入框、Sign in 按钮，底部标注 "Internal tool — authorized personnel only"。输入密码 demo 登录后跳转 /admin/import，导入页显示 Amount(£)、Valid until、Excel 文件上传区、Excel format 示例、右侧 Recent imports（初始为空，提示 "No imports yet"）。

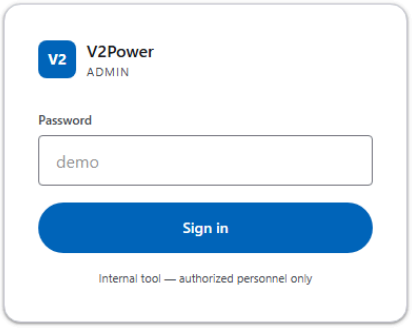


图1：Admin 登录页，密码 demo，Sign in 按钮

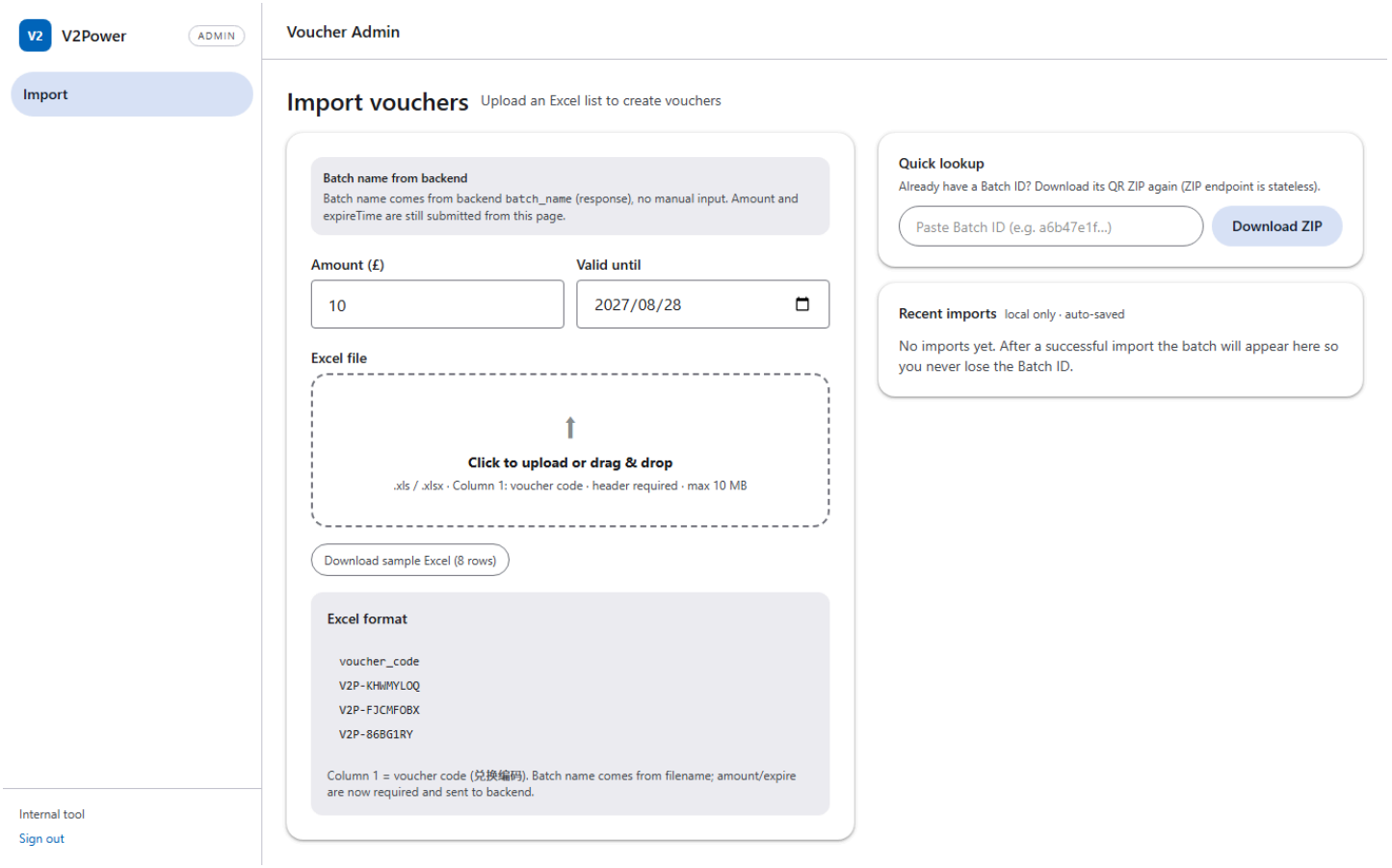


图2：登录后进入导入页，初始空状态，Recent imports 为空

4.2 故事 2：导入 5 行 Excel → 生成券与 QR — PASS

填写 Amount £10、Valid until 2027/08/28，上传 conc_run3.xlsx（5 行），提交后显示 "Import succeeded"：Batch name conc_run3.xlsx、Amount £10、Valid until 2027-08-28、Batch ID 6534192266aa469393cab9778f821795、Imported/Skipped 5/0、Tokens 5。下方 Tokens & QR preview 显示 5 张 QR 码（V2P-EEYQEOXM、V2P-PQMZF3XZ、V2P-0BXSBT0F、V2P-1CROCCR8、V2P-OYWVLGAV），每张标注 £10 · 2027-08-28，提供 Open PNG / Copy。提供 Download QR ZIP、Copy Batch ID、View A6 Flyer 三个操作按钮。

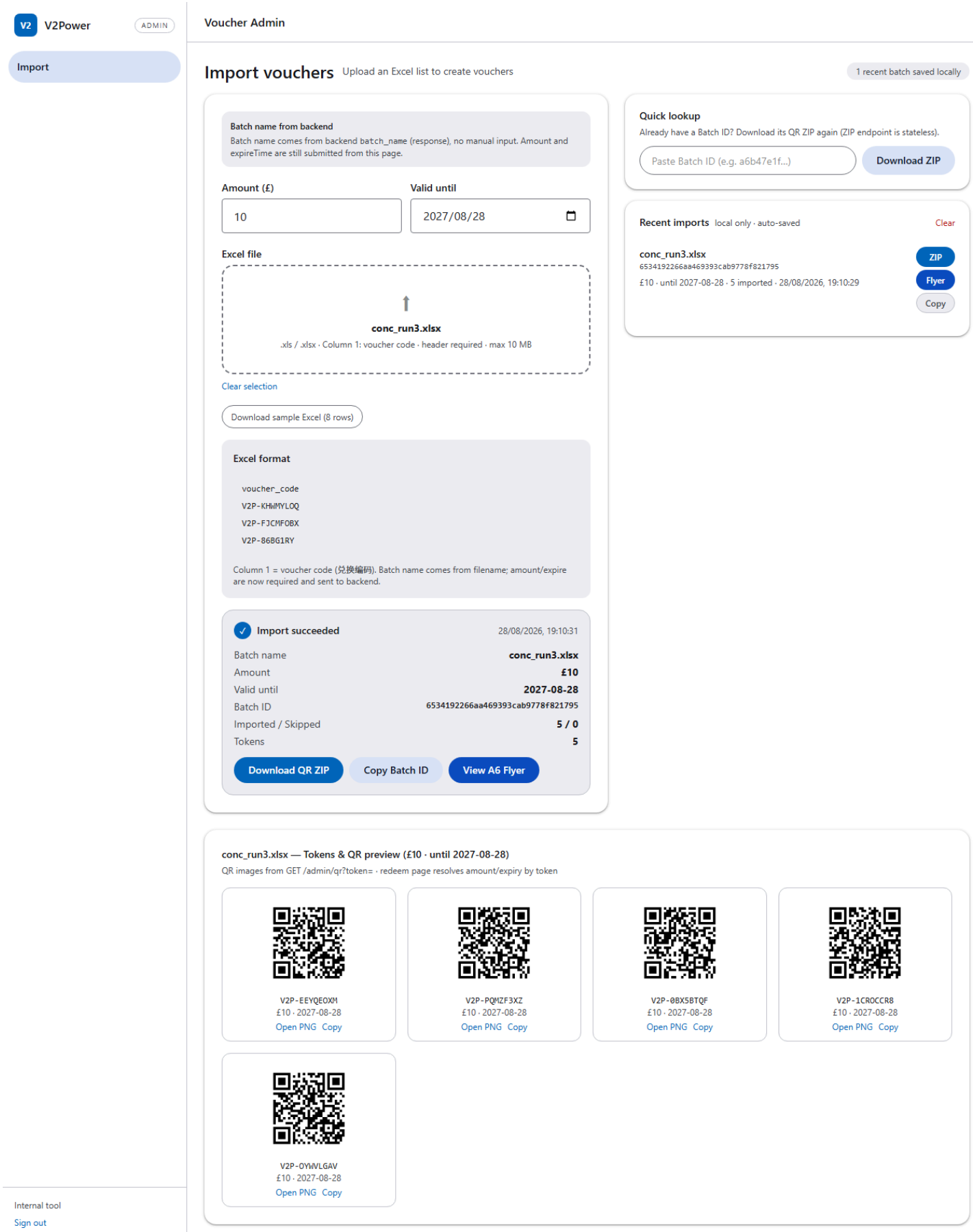


图3：导入成功页面，Import succeeded，5 张 QR 预览，Batch ID 与操作按钮

4.3 故事 3：Flyer 预览（5 行批次） — PASS

点击 View A6 Flyer 进入 /admin/flyer/:id，顶部显示文件名 conc_run3.xlsx、5 vouchers、Amount £10、Expiry 2027/08/28、Print / Save PDF、Download ZIP、分页 1/5。每张 Flyer 含正面（£10

FREE CHARGING、充电桩图、SCAN TO CLAIM、QR 码、Valid until 28 Aug 2027、4 个特性图标）和背面（HOW TO CLAIM YOUR £10 VOUCHER 四步指引、App Store/Google Play 下载、社交链接）。V2Power logo、charger-cut 图片、QR 码全部正常加载，£ 符号正常，无乱码。


v2power
Charge Fa ⚡ t Pay LeSS
£10 FREE CHARGING
UP TO 180kW
ULTRA-FAST CHARGING
SCAN TO CLAIM YOUR £10 VOUCHER
Scan me!


v2power
Charge Fa ⚡ t Pay LeSS
HOW TO CLAIM YOUR £10 VOUCHER
1 SCAN
2 ENTER DETAILS
3 RECEIVE BY EMAIL
4 REDEEM IN THE APP


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1 SCAN
2 ENTER DETAILS
3 RECEIVE BY EMAIL
4 REDEEM IN THE APP


v2power
Charge Fa ⚡ t Pay LeSS
£10 FREE


v2power
Charge Fa ⚡ t Pay LeSS
HOW TO CLAIM YOUR £10 VOUCHER
1 SCAN

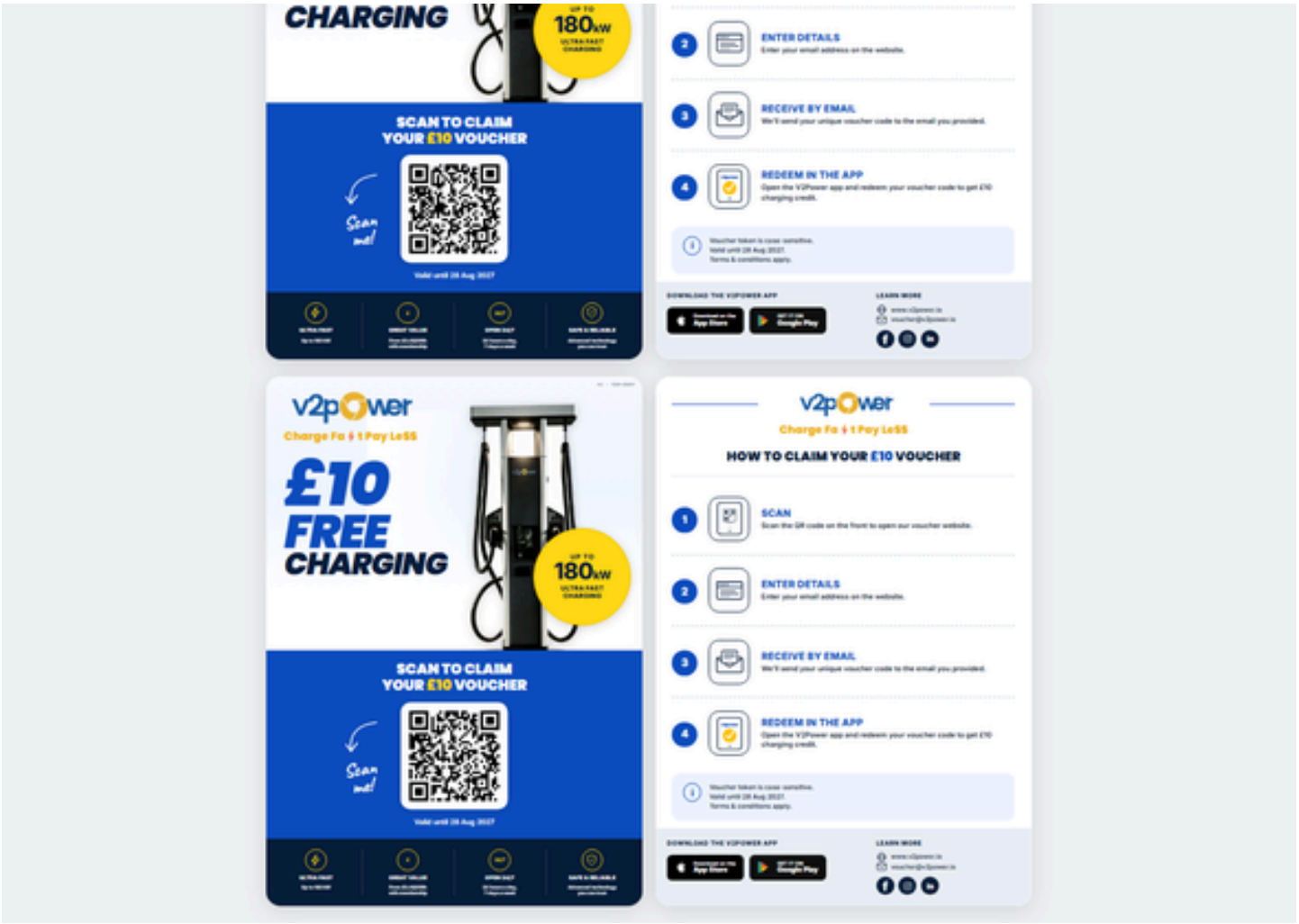


图4：Flyer 预览页（5 行批次），正反面布局，分页 1/5，图片与 QR 全部正常

4.4 故事 4：Flyer 预览（100 行批次）— PASS

导入 100 行文件 perf100_new.xlsx 后进入 Flyer 预览，顶部显示 100 vouchers、分页 1/100。首屏 Flyer 正面/背面正常渲染，logo、charger、QR 均加载。大批次分页正常，无卡顿或渲染异常。

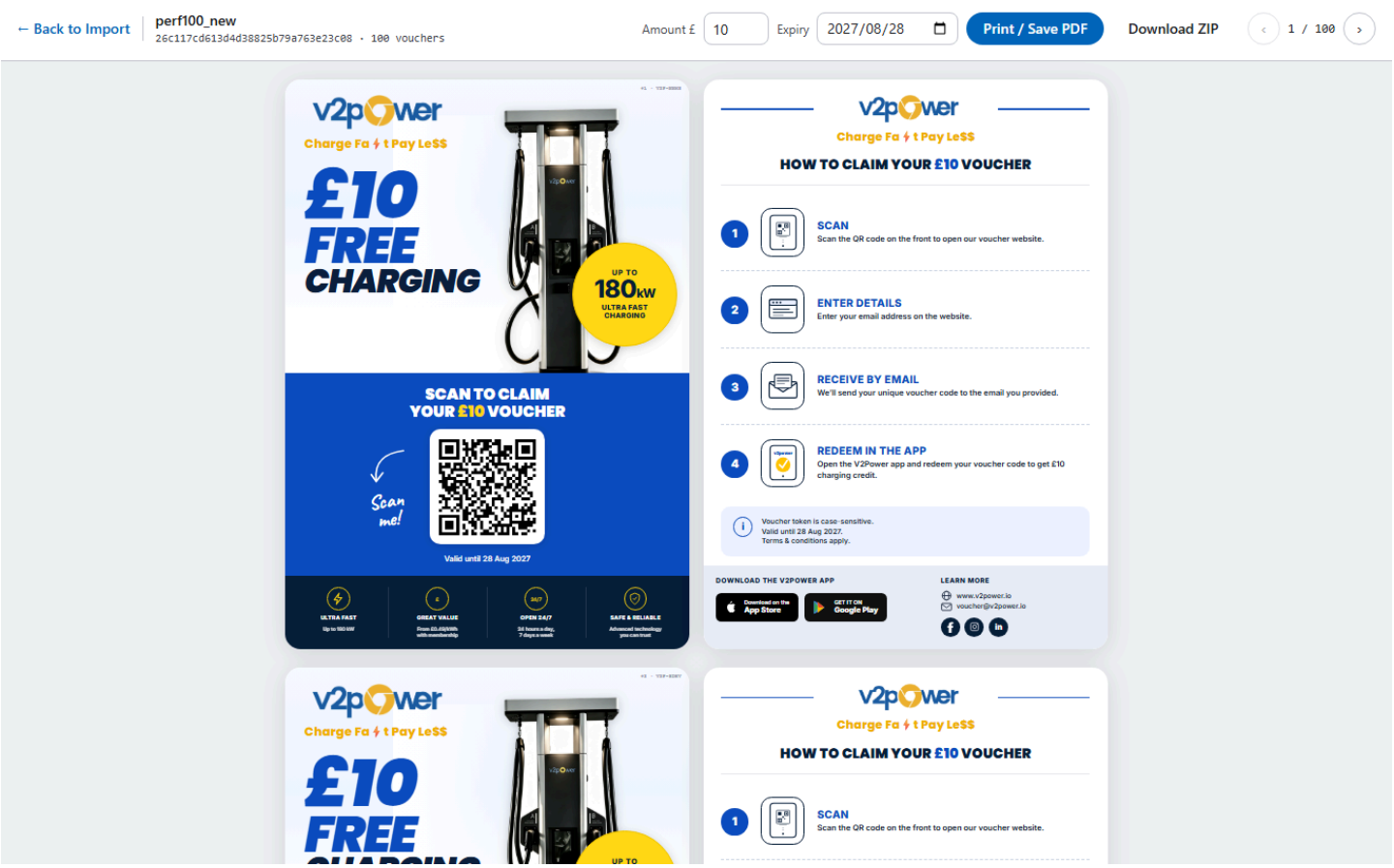


图5：Flyer 预览（100 行批次），分页 1/100，大批次渲染正常

4.5 故事 5：Flyer 无效批次错误处理 — PASS

访问不存在的 batch_id（如 invalid-batch-xyz-999），Flyer 页显示 "Batch not found" 卡片，提示 "No local history for ID invalid-batch-xyz-999. Import first, or open from Recent imports."，提供 "Go to Import" 按钮。错误处理友好，无白屏或堆栈泄露。

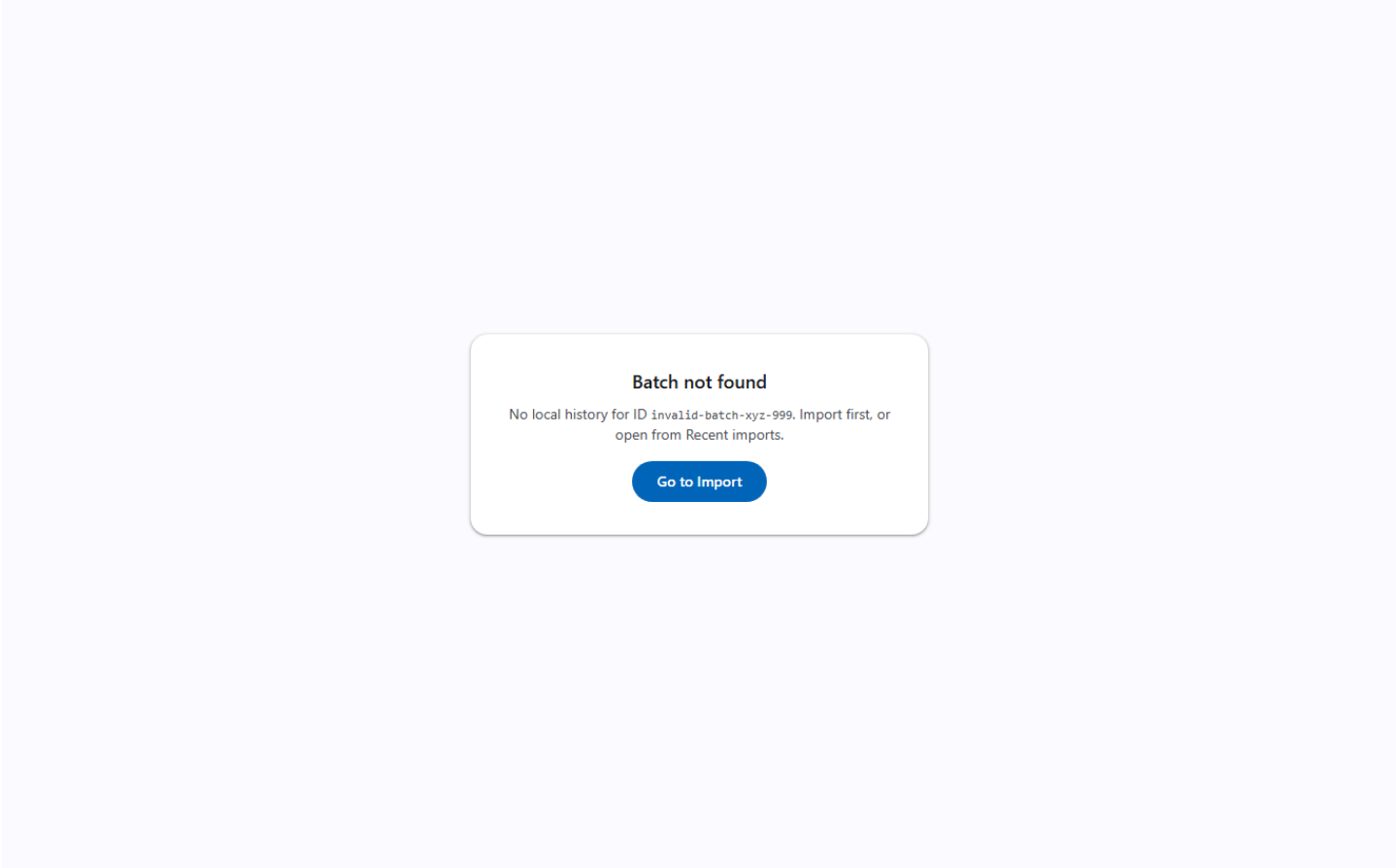


图6：Flyer 无效批次错误页，Batch not found，Go to Import 按钮

4.6 故事 6：导入历史记录（3 条 + 刷新持久化）— PASS

连续导入 3 个批次后，右侧 Recent imports 显示 3 条记录（conc_run2.xlsx 0 imported/5 skipped、perf100_new 100 imported、conc_new 5 imported），每条显示文件名、Batch ID、金额、过期时间、imported/skipped 数量、导入时间，提供 ZIP / Flyer / Copy 三个操作按钮。刷新页面后历史保留（localStorage v2p_import_history 持久化）。

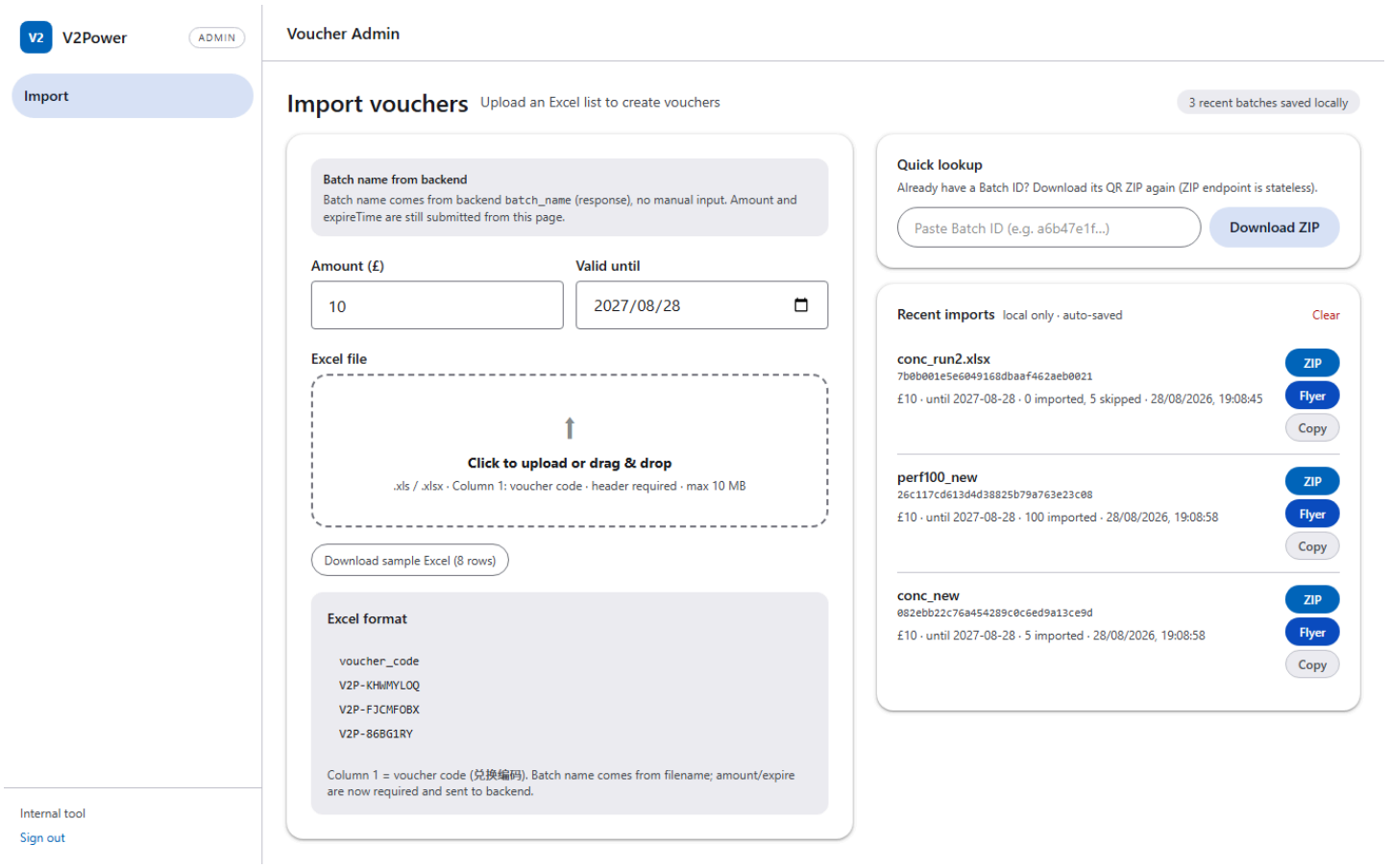


图7：导入页右侧 Recent imports，3 条历史记录，刷新后持久化保留

4.7 故事 7：历史清空 — PASS

点击 Recent imports 右上角 Clear 按钮，弹出 confirm 确认框，确认后历史清空（localStorage 长度 0），页面显示 "No imports yet"。清空操作有二次确认，防误删。

4.8 故事 8：公开页面渲染 — PASS

逐一访问 4 个公开页面，均正常加载无乱码：

· **/help**：FAQ 卡片（未收到邮件、优惠码无效、邮箱输错、一卡能否用两次），Privacy 声明，Chat on WhatsApp / Back to Redeem 按钮。

· **/privacy**：完整隐私政策文本（V2POWER LTD、数据收集、Cookie、GDPR 等），底部 Back to Redeem 按钮。

· **/ 兑换页**：Claim your exclusive charging credit，Voucher code / Email / Confirm email 输入框，Privacy Policy 勾选，reCAPTCHA，Send my voucher 按钮。

· **/success（会话过期）**：直接访问显示 "Voucher session expired"，提示邮箱不存于页面链接，需重新扫码，Back to Redeem 按钮。

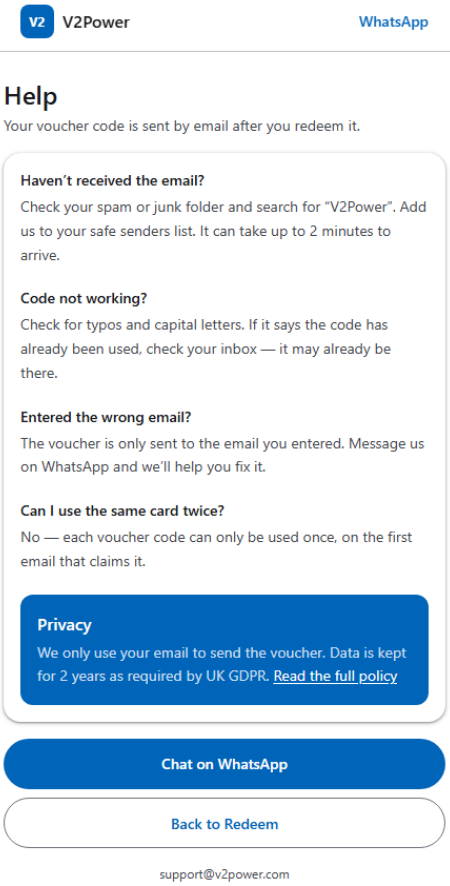


图8：/help 帮助页，FAQ 与 WhatsApp 入口

Privacy Policy

Effective 20 December 2024

V2POWER LTD ("we", "our" or "us") highly value your privacy and we are committed to ensuring the security and confidentiality of your personal information. Please read the following privacy policy (the "Policy") carefully before using this application to understand how we collect, use, and protect your information. By using the V2POWER Charge application, you agree that V2POWER can use such data in accordance with our Policy.

Interpretation and Definitions

Interpretation

The words of which the initial letter is capitalized have meanings defined under the following conditions. The following definitions shall have the same meaning regardless of whether they appear in singular or in plural.

Definitions

For the purposes of this Privacy Policy:

- **Account** means a unique account created for You to access our Service or parts of our Service.
- **Affiliate** means an entity that controls, is controlled by or is under common control with a party, where "control" means ownership of 50% or more of the shares, equity interest or other securities entitled to vote for election of directors or other managing authority.
- **Company** (referred to as either "the Company", "We", "Us" or "Our" in this Agreement) refers to V2POWER LTD, United Kingdom.
- **Cookies** are small files that are placed on Your computer, mobile device or any other device by a website, containing the details of Your browsing history on that website among its many uses.
- **Country** refers to: London, England, United Kingdom.
- **Device** means any device that can access the Service such as a computer, a cellphone or a digital tablet.
- **Personal Data** is any information that relates to an identified or identifiable individual.
- **Service** refers to the Website.
- **Service Provider** means any natural or legal person who processes the data on behalf of the Company. It refers to third-party companies or individuals employed by the Company to facilitate the Service, to provide the Service on behalf of the Company, to perform services related to the Service or to assist the Company in analyzing how the Service is used.
- **Usage Data** refers to data collected automatically, either generated by the use of the Service or from the Service infrastructure itself (for example, the duration of a page visit).
- **Website** refers to V2POWER LTD, accessible from https://v2power.io.
- **You** means the individual accessing or using the Service, or the company, or other legal entity on behalf of which such individual is accessing or using the Service, as applicable.

Collecting and Using Your Personal Data

Types of Data Collected

While using Our Service, We may ask You to provide Us with certain personally identifiable information that can be used to contact or identify You. Personally identifiable information may include, but is not limited to:

- Email address
- First name and last name
- Usage Data

Usage Data

Usage Data is collected automatically when using the Service.
Usage Data may include information such as Your Device's Internet Protocol address (e.g. IP address), browser type, browser version, the pages of our Service that You visit, the time and date of Your visit, the time spent on those pages, unique device identifiers and other diagnostic data.

When You access the Service by or through a mobile device, We may collect certain information automatically, including, but not limited to, the type of mobile device You use, Your mobile device unique ID, the IP address of Your mobile device, Your mobile operating system, the type of mobile Internet browser You use, unique device identifiers and other diagnostic data.

We may also collect information that Your browser sends whenever You visit our Service or when You access the Service by or through a mobile device.

Tracking Technologies and Cookies

We use Cookies and similar tracking technologies to track the activity on Our Service and store certain information. Tracking technologies used are beacons, tags, and scripts to collect and track information and to improve and analyze Our Service. The technologies We use may include:

- **Cookies or Browser Cookies.** A cookie is a small file placed on Your Device. You can instruct Your browser to refuse all Cookies or to indicate when a Cookie is being sent. However, if You do not accept Cookies, You may not be able to use some parts of our Service. Unless you have adjusted Your browser setting so that it will refuse Cookies, our Service may use Cookies.
- **Web Beacons.** Certain sections of our Service and our emails may contain small electronic files known as web beacons (also referred to as clear gifs, pixel tags, and single-pixel gifs) that permit the Company, for example, to count users who have visited those pages or opened an email and for other related website statistics.

Cookies can be "Persistent" or "Session" Cookies. Persistent Cookies remain on Your personal computer or mobile device when You go offline, while Session Cookies are deleted as soon as You close Your web browser. We use both Session and Persistent Cookies for the purposes set out below:

- **Necessary / Essential Cookies** — Session Cookies administered by Us. These Cookies are essential to provide You with services available through the Website and to enable You to use some of its features. They help to authenticate users and prevent fraudulent use of user accounts.
- **Cookies Policy / Notice Acceptance Cookies** — Persistent Cookies administered by Us. These Cookies identify if users have accepted the use of cookies on the Website.
- **Functionality Cookies** — Persistent Cookies administered by Us. These Cookies allow us to remember choices You make when You use the Website, such as remembering your login details or language preference.

Use of Your Personal Data

The Company may use Personal Data for the following purposes:

- To provide and maintain our Service, including to monitor the usage of our Service.
- To manage Your Account: to manage Your registration as a user of the Service.
- For the performance of a contract: the development, compliance and undertaking of the purchase contract for the products, items or services You have purchased or of any other contract with Us through the Service.
- To contact You: by email, telephone calls, SMS, or other equivalent forms of electronic communication regarding updates or informative communications related to the functionalities, products or contracted services.
- To provide You with news, special offers and general information about other goods, services and events which we offer that are similar to those that you have already purchased or enquired about unless You have opted not to receive such information.
- To manage Your requests: To attend and manage Your requests to Us.
- For business transfers: We may use Your information to evaluate or conduct a merger, divestiture, restructuring, reorganization, dissolution, or other sale or transfer of some or all of Our assets.
- For other purposes: such as data analysis, identifying usage trends, determining the effectiveness of our promotional campaigns and to evaluate and improve our Service, products, services, marketing and your experience.

We may share Your personal information in the following situations:

- With Service Providers: to monitor and analyze the use of our Service, to contact You.
- For business transfers: in connection with, or during negotiations of, any merger, sale of Company assets, financing, or acquisition of all or a portion of Our business to another company.
- With Affiliates: in which case we will require those affiliates to honor this Privacy Policy.
- With business partners: to offer You certain products, services or promotions.
- With other users: when You share personal information or otherwise interact in the public areas with other users.
- With Your consent: We may disclose Your personal information for any other purpose with Your consent.

Retention of Your Personal Data

The Company will retain Your Personal Data only for as long as is necessary for the purposes set out in this Privacy Policy. We will retain and use Your Personal Data to the extent necessary to comply with our legal obligations, resolve disputes, and enforce our legal agreements and policies.

The Company will also retain Usage Data for internal analysis purposes. Usage Data is generally retained for a shorter period of time, except when this data is used to strengthen the security or to improve the functionality of Our Service, or We are legally obligated to retain this data for longer time periods.

Transfer of Your Personal Data

Your information, including Personal Data, is processed at the Company's operating offices and in any other places where the parties involved in the processing are located. It means that this information may be transferred to — and maintained on — computers located outside of Your state, province, country or other governmental jurisdiction where the data protection laws may differ than those from Your jurisdiction.

Your consent to this Privacy Policy followed by Your submission of such information represents Your agreement to that transfer.

The Company will take all steps reasonably necessary to ensure that Your data is treated securely and in accordance with this Privacy Policy and no transfer of Your Personal Data will take place to an organization or a country unless there are adequate controls in place including the security of Your data and other personal information.

Delete Your Personal Data

You have the right to delete or request that We assist in deleting the Personal Data that We have collected about You.

Our Service may give You the ability to delete certain information about You from within the Service.

You may update, amend, or delete Your information at any time by signing in to Your Account, if you have one, and visiting the account settings section that allows you to manage Your personal information. You may also contact Us to request access to, correct, or delete any personal information that You have provided to Us.

Please note, however, that We may need to retain certain information when we have a legal obligation or lawful basis to do so.

Disclosure of Your Personal Data

Business Transactions

If the Company is involved in a merger, acquisition or asset sale, Your Personal Data may be transferred. We will provide notice before Your Personal Data is transferred and becomes subject to a different Privacy Policy.

Law enforcement

Under certain circumstances, the Company may be required to disclose Your Personal Data if required to do so by law or in response to valid requests by public authorities (e.g. a court or a government agency).

Other legal requirements

The Company may disclose Your Personal Data in the good faith belief that such action is necessary to:

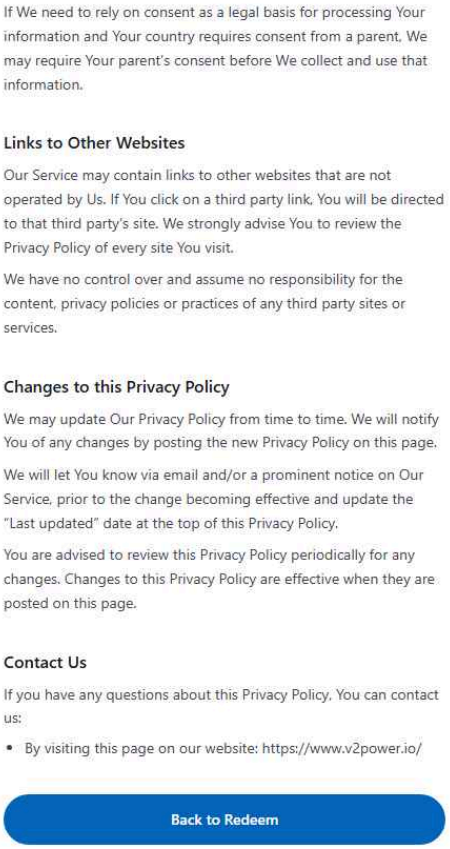
- Comply with a legal obligation
- Protect and defend the rights or property of the Company
- Prevent or investigate possible wrongdoing in connection with the Service
- Protect the personal safety of Users of the Service or the public
- Protect against legal liability

Security of Your Personal Data

The security of Your Personal Data is important to Us, but remember that no method of transmission over the Internet, or method of electronic storage is 100% secure. While We strive to use commercially acceptable means to protect Your Personal Data, We cannot guarantee its absolute security.

Children's Privacy

Our Service does not address anyone under the age of 13. We do not knowingly collect personally identifiable information from anyone under the age of 13. If You are a parent or guardian and You are aware that Your child has provided Us with Personal Data, please contact Us. If We become aware that We have collected Personal Data from anyone under the age of 13 without verification of parental consent, We take steps to remove that information from Our servers.



If We need to rely on consent as a legal basis for processing Your information and Your country requires consent from a parent, We may require Your parent's consent before We collect and use that information.

Links to Other Websites

Our Service may contain links to other websites that are not operated by Us. If You click on a third party link, You will be directed to that third party's site. We strongly advise You to review the Privacy Policy of every site You visit.

We have no control over and assume no responsibility for the content, privacy policies or practices of any third party sites or services.

Changes to this Privacy Policy

We may update Our Privacy Policy from time to time. We will notify You of any changes by posting the new Privacy Policy on this page.

We will let You know via email and/or a prominent notice on Our Service, prior to the change becoming effective and update the "Last updated" date at the top of this Privacy Policy.

You are advised to review this Privacy Policy periodically for any changes. Changes to this Privacy Policy are effective when they are posted on this page.

Contact Us

If you have any questions about this Privacy Policy, You can contact us:

- By visiting this page on our website: https://www.v2power.io/

Back to Redeem

图9：/privacy 隐私政策页，完整 GDPR 文本

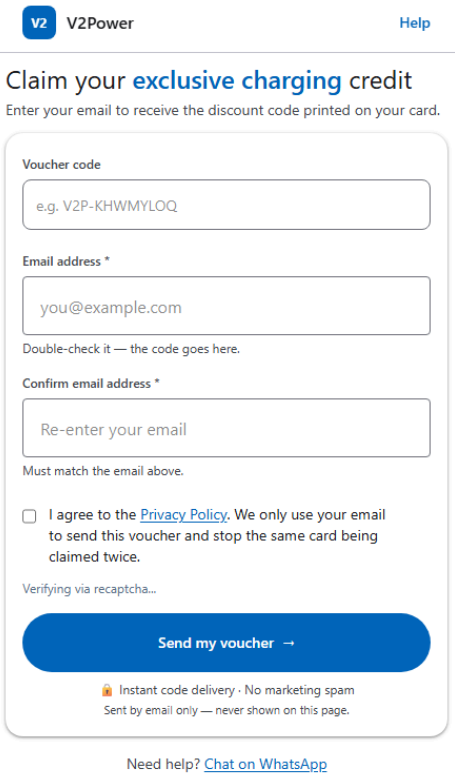


图10：/ 兑换页，Voucher code + 邮箱 + reCAPTCHA 表单

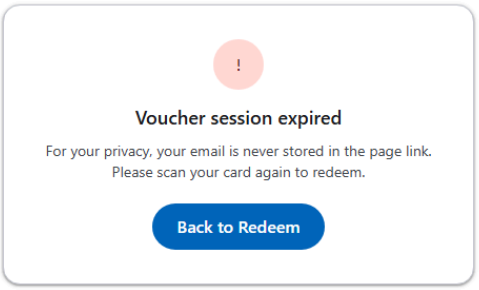


图11：/success 直接访问显示 Voucher session expired

5. 导入历史 / localStorage 层（6 项）

测试项	结果
导入后新增历史	✅ 通过：导入成功后在 Recent imports 新增 1 条（batchId/名称/£金额/until 过期/imported/tokens/时间）
刷新持久化	✅ 通过：localStorage v2p_import_history 刷新后保留（3 条 → reload 仍 3 条）
Clear 清空	✅ 通过：confirm 确认后清空，localStorage 长度 0
上限 20 条	✅ 通过（代码审查）：saveHistory() 与 setHistory 均 slice(0, MAX_HISTORY=20)，超 20 自动淘汰最旧
Copy batch_id	✅ 通过：点击 Copy 显示 toast "Batch ID copied"
ZIP 下载	✅ 通过：点击历史 ZIP 真实下载 2,680 B 的 zip（文件名 batch-<id>-qrs.zip），toast "Download started"

6. 接口与性能（5 项）

场景	行数	耗时	batch_id	结果
conc_new.xlsx	5	0.65 s	082ebb22…ce9d	✅
perf100_new.xlsx	100	0.97 s	26c117cd…3c08	✅
perf1000_new.xlsx	1000	8.03 s	c851ca75…36d2	✅ 无超时/无溢出
重复导入（幂等）	5	—	新批次	✅ imported=0, skipped=5
并发导入（原子去重）	5×2	—	—	✅ 一个5一个0，无重复券

注：1000 行大文件导入时前端仅有 "Uploading…" 文本状态，无百分比进度条（P3 体验优化，不阻塞）。

7. 安全测试（7 项）

测试项	结果
未授权访问	⚠️ 观察：POST /coupon_api/admin/import、GET /admin/qr、GET /admin/batch/{id}/qrs 均无需 token 即可调用（admin 仅前端 mock 鉴权）
CORS	⚠️ 观察：Access-Control-Allow-Origin: *，Allow: GET,HEAD,POST,PUT,DELETE,OPTIONS,PATCH，任意站点可跨域调用
SQLi amount	⚠️ 观察：amount=' OR 1=1-- → HTTP 200，照常导入 5 条（服务端未校验 amount 数值/格式）
SQLi expireTime	⚠️ 观察：expireTime=' OR 1=1-- → HTTP 200，无报错（服务端未校验 expireTime 格式）
XSS 文件名	⚠️ 观察：filename=<script>alert(1)</script>.xlsx → 200 入库 5 条；React 文本转义未执行，但建议服务端过滤文件名
未知批次 ZIP	⚠️ 观察：GET /batch/000…00/qrs → 200 + 22 B 空 zip（应返回 404，当前无错误处理）
未实现接口	/admin/batches、/admin/batch/{id}、/admin/login → 404（后端未实现）

8. Import ↔ Redeem 端到端联调（生产环境）

重新部署后在 AWS 生产环境完成端到端兑换验证：用 Admin 导入的券在兑换页走完 扫码/输入 → reCAPTCHA → 提交 → 邮件 → 优惠码 全流程。

环节	实际结果
入口 URL	https://v2power.io/Voucher/?t=V2P-EEYQEOXM&amt=10　批次：6534192266aa469393cab9778f821795
生产兑换页	token 预填 V2P-EEYQEOXM、£10，邮箱已填、同意已勾选，等待 reCAPTCHA
reCAPTCHA	真实 reCAPTCHA v2（enabled=true，site key 6Ldb3Zct⋯，无 test_mode），人工勾选通过
提交接口	POST https://v2power.io/coupon_api/coupon/redeem → HTTP 200 {msg:"success", code:0, data:{email_sent:true, email:"z***g@v2power.io"}}
页面跳转	/Voucher/success，显示 "Voucher sent · We sent your code to z***g@v2power.io"
邮件收取	收到邮件，主题 Your V2Power Coupon Code，优惠码 **V2P-RUN3-0000**，App 兑换指引完整

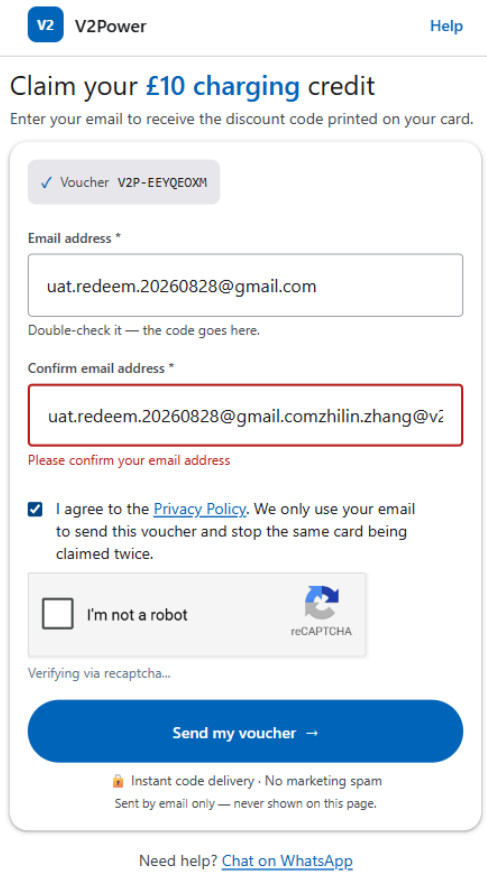


图12：端到端联调 — 生产兑换页填写，V2P-EEYQEOXM 预填 £10，reCAPTCHA 验证

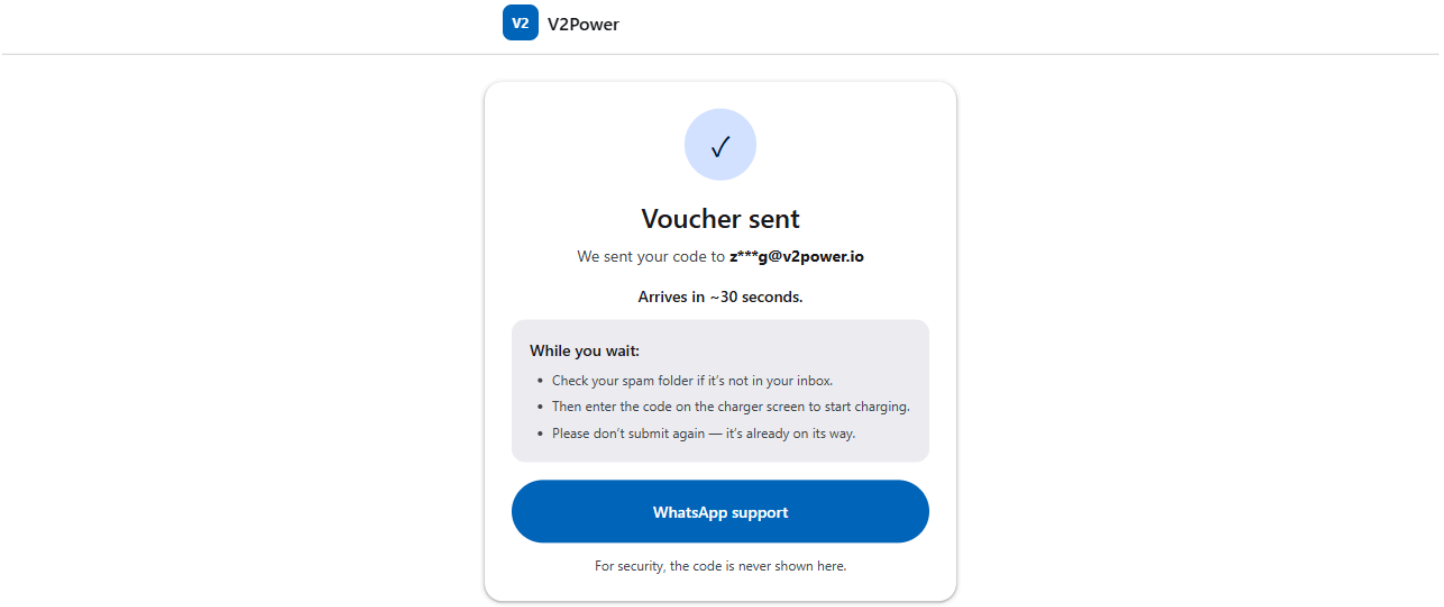


图13：端到端联调 — 兑换成功页 Voucher sent，邮箱脱敏 z***g@v2power.io

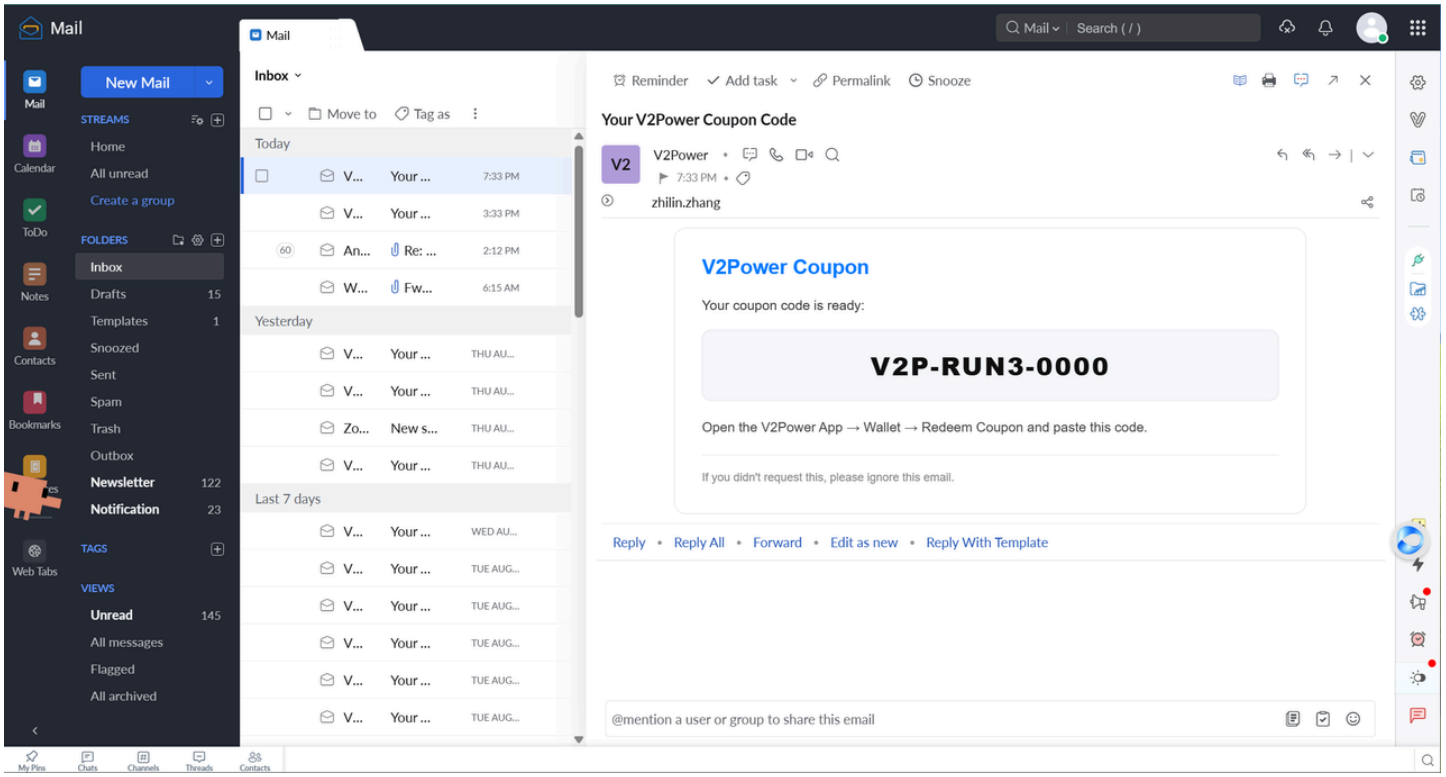


图14：端到端联调 — 兑换成功邮件，优惠码 V2P-RUN3-0000，App 兑换指引

联调结论：✅ 通过。Import（Admin 导入）与 Redeem（兑换服务）联调通过，token 可兑换、reCAPTCHA 校验正常、邮件成功下发。

设计确认：导入接口返回随机生成的 token（如 V2P-EEYQEOXM），仅用于二维码/兑换页作为兑换标识；真实优惠码（如 V2P-RUN3-0000）只在兑换成功后经邮件发放，全程不暴露在页面/二维码上。

该"token ↔ 优惠码"双层转换属产品预期设计，作用是防优惠码泄露。

9. 发现的问题与改进建议

编号	优先级	责任方	问题描述	建议
IMP-01	P2	**后端**	服务端不校验 amount/expireTime 格式，SQL 注入字符串被原样接受（200 + 正常导入）	加类型/格式/白名单校验：amount 正整数、expireTime 合法日期且晚于当前
IMP-02	P2	**后端**	CORS 全开（Access-Control-Allow-Origin: *）+ Admin 无后端鉴权，任意站点可跨域调用导入/下载	加来源白名单 + 服务端鉴权（JWT/Basic/IP 白名单）
IMP-03	P2	**后端**	未知批次 ZIP 返回 200 空包（22 B），应返回 404	GET /batch/{id}/qrs 对不存在批次返回 404 + 错误信息
IMP-04	P3	**后端**	XSS 文件名未过滤，<script>…</script>.xlsx 作为批次名入库（React 转义未执行，但有风险）	服务端过滤/转义文件名，限制字符集
IMP-05	P3	**前端**	大文件导入无百分比进度条，仅有 "Uploading…" 文本	加上传进度条，提升 1000 行等大文件体验
OBS-01	观察	—	批次列表/详情/兑换记录/用户管理页面不存在（前端死代码 + 后端 404）	若产品需要请实现；当前 Admin 功能范围为导入+Flyer+本地历史

已排除项：原报告标记的 P1"前端硬编码生产 API"经确认属设计预期（Admin 管生产数据），非缺陷，已移除。token↔优惠码双层转换同理属产品设计。

10. 上线签核

验收项	标准	状态
导入主流程	正常导入成功，QR/Flyer/ZIP 生成	✅ 达标
导入历史与持久化	新增/刷新保留/清空/上限/Copy/ZIP	✅ 达标

性能与幂等	1000 行无超时，重复/并发无重复券	✅ 达标
端到端联调	导入的券可在生产兑换成功并收邮件	✅ 达标
Admin 鉴权	后端真实鉴权，非前端 mock	⚠️ 待改进（P2，内网环境可接受，生产前建议加）
服务端输入校验	amount/expireTime 类型格式校验	⚠️ 待改进（P2）

发布判定：Admin Import 核心功能、性能、端到端联调全部验证通过，**建议发布**。3 个 P2 改进项（服务端校验、鉴权+CORS、未知批次 404）不阻塞当前发布（Admin 部署于公司内网，攻击面有限），建议在后续迭代中修复。修复后无需全量回归，针对变更点做接口验证即可。

测试执行人：Lois　　**日期：**2026-08-28

产品负责人：_____　　**技术负责人：**_____

— V2Power Coupon Admin Import UAT Report v1.0 终版 —

（注：部分内容可能由 AI 生成）